

Multimodal Emotion Recognition in Conversations via Class-Wise Adaptive Modality Fusion and Affective Geometry

Oriol Marín

Roger Marí

Gloria Haro

Rafael Redondo

Motivation

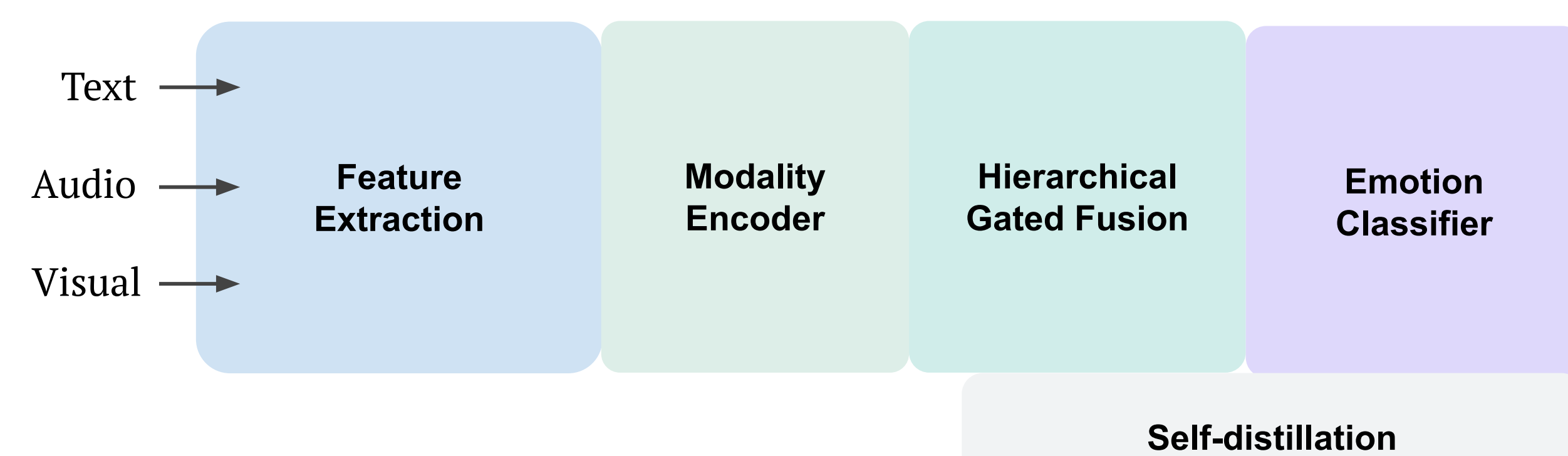
Emotion Recognition in Conversations (ERC) must jointly model heterogeneous text, audio, and visual cues while tracking how affect evolves across a dialogue.

Three limitations remain open:

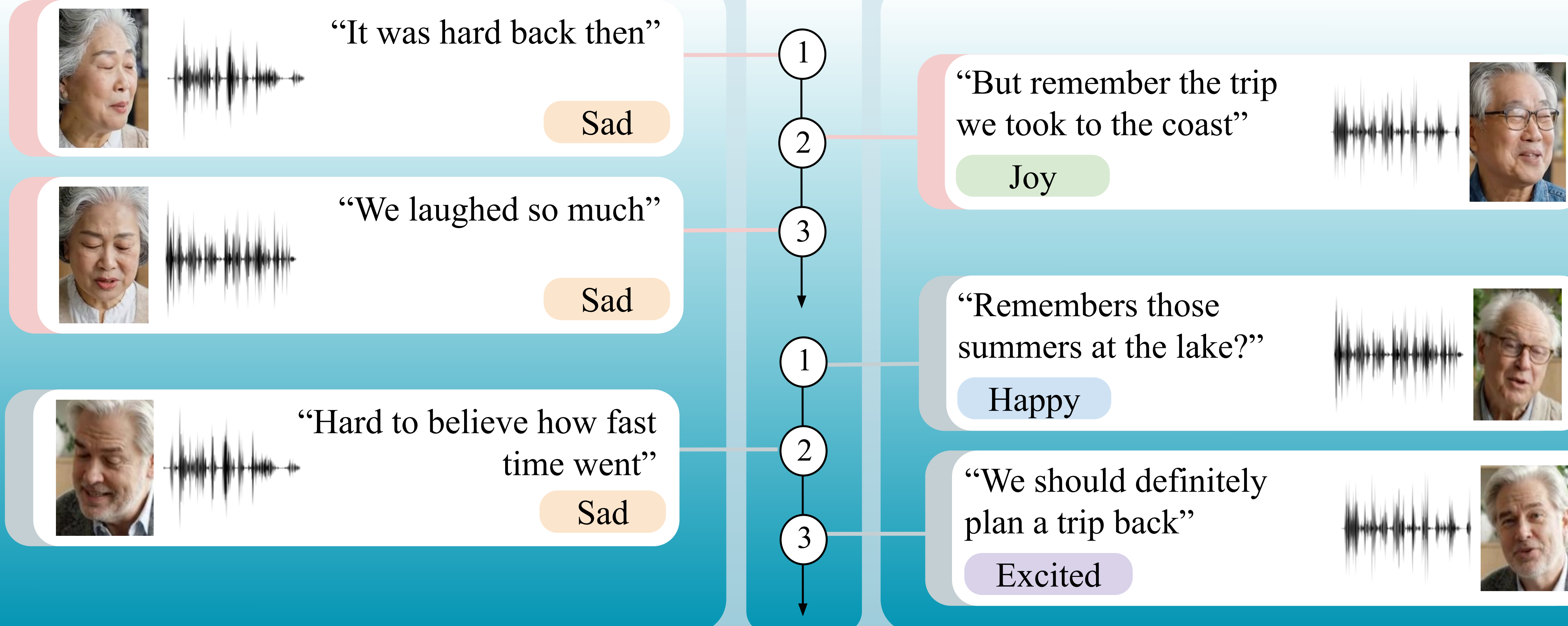
1. Standard fusion is dominated by text.
2. Modality informativeness varies across utterances and emotion categories.
3. Discrete labels ignore affective geometry, making emotion-shift recognition hard.

Method Overview

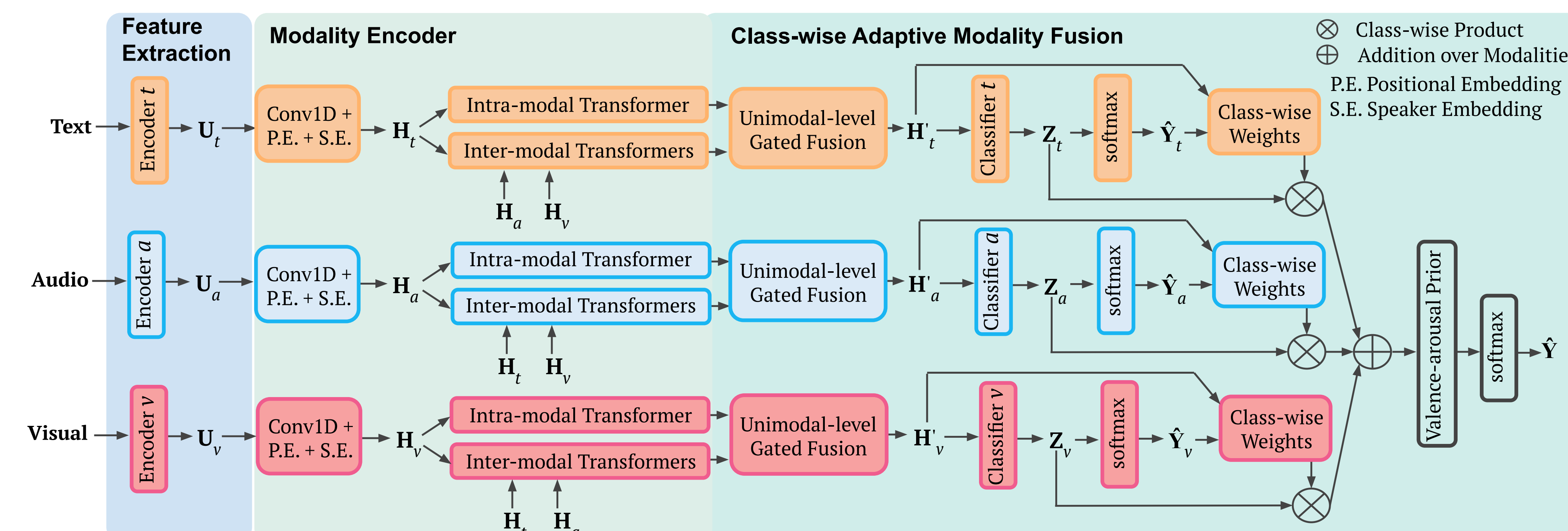
We build on the Self-Distillation Transformer (SDT) [1].



Text, audio, and visual features are encoded with intra- and inter-modal Transformers. We keep this backbone and introduce **class-wise adaptive fusion** and a **valence-arousal prior**.

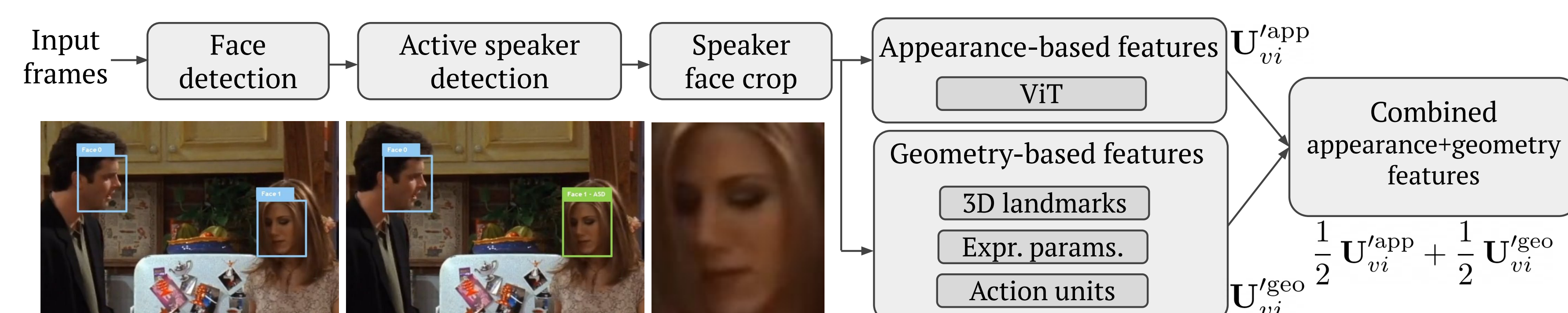


2. Class-Wise Adaptive Fusion



Our class-wise fusion predicts weights per modality and emotion class, and applies them at the logit level, allowing modality contributions to vary according to their informativeness for each emotion class.

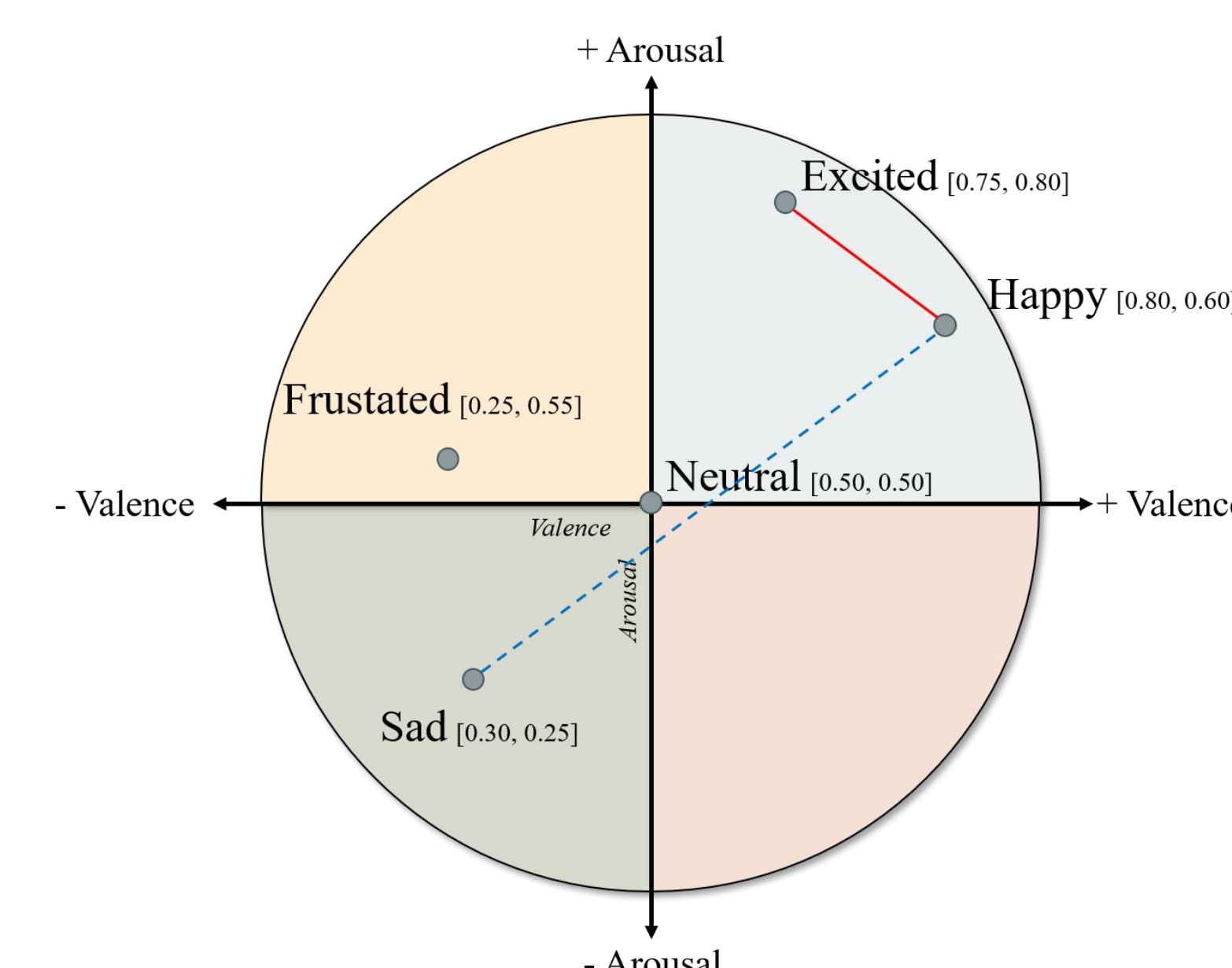
1. Visual Stream: Appearance + Geometry



A ViT encodes **appearance**; a small MLP encodes **geometry** (3D landmarks, expression parameters, or action units). Both feature types are projected to a shared space and combined into an enhanced visual representation.

3. Valence-Arousal Prior

Happy → excited is affectively closer than happy → sad. We bias predictions toward plausible affective transitions [2].



1. Expected state:

$$\hat{\mathbf{v}}_i = \sum_{c=1}^C \hat{\mathbf{p}}_i^c \cdot \mathbf{v}_c,$$

2. Shift magnitude:

$$\beta_i = \|\hat{\mathbf{v}}_i - \hat{\mathbf{v}}_{i-1}\|$$

3. Prior correction:

$$\hat{\mathbf{z}}'_i = \hat{\mathbf{z}}_i + \alpha \cdot \beta_i \cdot \phi_i,$$

Datasets

MELD [3] - 13,708 utterances, 1,433 multi-party TV dialogues, 7 emotions. In-the-wild recording conditions.

IEMOCAP [4] - 7,433 utterances, dyadic lab sessions, 6 emotions. Controlled setting with continuous valence-arousal labels available.

Results

Model	MELD	IEMOCAP	Mean
Original SDT [1]	66.60	74.08	70.34
Updated baseline	75.49	69.50	72.50
+ Appearance + geometry	75.76	73.86	74.81
+ Class-wise fusion	75.93	74.11	75.02

Best in red (weighted-F1 (%)).

Emotion-shift accuracy gain with valence-arousal prior

	Shifted	Stable
MELD	+0.30	+0.11
IEMOCAP	+0.74	0.00

Best in red (accuracy (%)).

- Geometry-enhanced visuals complement appearance features, especially under controlled conditions (IEMOCAP).
- Class-wise fusion outperforms the baseline SDT softmax fusion gate using 3x fewer parameters.
- Valence-arousal prior improves shifted turns while preserving performance on stable ones.

Try it yourself!

<https://github.com/multimedia-eurecat/classwise-multimodal-ERC>



References

- [1] Ma et al. A transformer-based model with self-distillation for multimodal emotion recognition in conversations. *IEEE TMM*, 2024.
- [2] Russell. A circumplex model of affect. *Journal of Personality and Social Psychology*, 1980.
- [3] Poria et al. MELD: A multimodal multi-party dataset for emotion recognition in conversations. *ACL*, 2019.
- [4] Busso et al. IEMOCAP: Interactive emotional dyadic motion capture database. *LREC*, 2008.